GRADE 10 / MYP 5 • FORCE & MOTION

Defining & quantifying motion

Lesson 2 • Describe a car’s motion with carefully defined quantities, equations and SI units.

LEARNING GOALS

By the end, you should be able to...

- Distinguish position, distance and displacement.
- Calculate average speed, average velocity and acceleration.
- Use signed quantities along a straight road.
- Show a clear equation, substitutions and units.
- Solve problems involving uniformly accelerated motion.

QUICK RECAP

Two journeys can end in the same place

DISTANCE

How much ground was covered along the entire path?

A scalar (m).

DISPLACEMENT

Final position − initial position.

Includes direction (m).

EQUATIONS

Position and displacement

$$\Delta x = x_f - x_i$$

Choose an origin, a positive direction and units.

From −4 m to +10 m: Δx = +14 m.

EQUATIONS

Average speed

average speed = total distance / Δt

One overall rate for the complete observed journey. Units: m/s.

A car travels 30 m in 6 s → average speed = 5 m/s.

EQUATIONS

Average velocity

$$v_{avg} = \Delta x / \Delta t$$

Uses signed displacement, not total distance. Units: m/s.

From +12 m to −8 m in 5 s → v_{avg} = −4 m/s.

EQUATIONS

Acceleration

$$a_{avg} = (v_f - v_i) / \Delta t$$

Velocity can change in size, direction, or both. Units: m/s^2.

From +2 m/s to +10 m/s in 4 s → +2 m/s^2.

SIGNS MATTER

Velocity versus acceleration

MOVING RIGHT; SLOWING

Velocity points right (+).
Acceleration points left (−).

MOVING LEFT; SPEEDING UP

Velocity points left (−).
Acceleration also points left (−).

EQUATIONS

Uniform acceleration

$$v_{avg} = (v_i + v_f) / 2$$

ONLY for constant acceleration in 1D.

From 0 to +8 m/s over 4 s → v_{avg} = +4 m/s.

EQUATIONS

Displacement under uniform acceleration

$$\Delta x = v_{avg} \times \Delta t$$

The mean of v_i and v_f applies when acceleration is constant.

Journey 3: (+4 m/s) × (4 s) = +16 m.

SIX STRAIGHT-LINE JOURNEYS

Describe first; calculate second

For each motion, determine displacement from start to finish, distance travelled, average speed, average velocity and acceleration.

- 1: Constant forward: 30 m in 6 s.
- 2: Constant backward: x +12 m → −8 m in 5 s.
- 3: From rest to +8 m/s in 4 s (uniform a).
- 4: From +8 m/s to rest in 4 s (uniform a).

TWO MORE JOURNEYS

Check the full interval

For each motion, determine displacement from start to finish, distance travelled, average speed, average velocity and acceleration.

- 5: +10 m/s to +6 m/s in 4 s (uniform a).
- 6: Starts +4 m/s; acceleration −2 m/s^2 for 4 s.
- Which journey ends where it started?
- Which two have the same distance and time?

WORKED CALCULATIONS

Journeys 3 and 4

JOURNEY 3

$a = (8 - 0)/4 = +2\ m/s^2$

$v_{avg} = 4\ m/s$

$\Delta x = +16\ m$

JOURNEY 4

$a = (0 - 8)/4 = -2\ m/s^2$

$v_{avg} = 4\ m/s$

$\Delta x = +16\ m$

GOOD SCIENTIFIC WORKING

Four steps

- 1 Write the equation.
- 2 Rearrange it if needed.
- 3 Substitute values with units.
- 4 State the answer, unit and direction if relevant.

PRACTICE & DISCUSSION

Work through the worksheet

- Start with MCQs 1–8.
- Then tackle the numbered free-response questions.
- Provide clear, logical workings for free-response questions.
- Extension: explore the other constant-acceleration equations.

LESSON SUMMARY

What you should understand

- Distance and displacement are not interchangeable.
- Average speed and average velocity can be different.
- Acceleration is a change in velocity per second.
- Signs, units and assumptions matter in every calculation.